Supplementary Table S1. Search strategy

Pubmed	
#1	((("Cough"[Mesh]) OR (cough[TIAB] OR coughing[TIAB] OR coughs[TIAB])) OR (("Bronchitis"[Mesh:NoExp]) OR "Bronchitis, Chronic"[Mesh])) OR (bronchitis[TIAB] OR bronchitic[TIAB])
#2	("Rhinitis"[Mesh] OR "Rhinitis, Allergic"[Mesh] OR "Sinusitis"[Mesh]) OR (rhinitis[tiab] OR sinusitis[tiab] OR rhinosinusitis[tiab])
#3	"Asthma"[Mesh] OR asthma[tiab] OR asthmatic[tiab] OR asthmatics[tiab]
#4	#1 OR #2 OR #3
#5	"Histamine Antagonists"[Mesh:NoExp] OR "Histamine H1 Antagonists"[Mesh] OR "Cetirizine"[Mesh] OR "fexofenadine" [Supplementary Concept] OR "Loratadine"[Mesh] OR "acrivastine"[Supplementary Concept] OR "epinastine" [Supplementary Concept] OR "ebastine"[Supplementary Concept] OR "azelastine" [Supplementary Concept] OR "rupatadine" [Supplementary Concept] OR "desloratadine" [Supplementary Concept] OR "bepotastine"[Supplementary Concept] OR "bilastine"[Supplementary Concept] OR "levocetirizine"[Supplementary Concept] OR "Olopatadine Hydrochloride"[Mesh] OR "levocabastine" [Supplementary Concept] OR "mizolastine" [Supplementary Concept] OR "alcaftadine" [Supplementary Concept]
#6	"Histamine H1 Antagonist"[tiab] OR "Histamine H1 receptor Antagonist"[tiab] OR "H1RA"[tiab] OR Cetirizine[tiab] OR Fexofenadine[tiab] OR Loratadine[tiab] OR Acrivastine[tiab] OR Epinastine[tiab] OR Ebastine[tiab] OR Azelastine[tiab] OR Rupatadine[tiab] OR Desloratadine[tiab] OR Bilastine[tiab] OR Levocetirizine[tiab] OR Olopatadine[tiab] OR levocabastine[tiab] OR mizolastine[tiab] OR alcaftadine[tiab]
#7	#5 OR #6
#8	#4 AND #7
#9	#8 AND (groups[tiab] OR trial[TIAB] OR randomly[TIAB] OR "drug therapy"[SH] OR placebo[TIAB] OR randomized[TIAB] OR "controlled clinical trial"[PT] OR "randomized controlled trial"[PT]) NOT (animals[MH] NOT (humans[MH] AND animals[MH]))
Embase	
#1	'coughing'/exp OR cough:ab,ti OR coughing:ab,ti OR coughs:ab,ti OR 'bronchitis'/de OR 'chronic bronchitis'/exp OR 'laryngotracheobronchitis'/exp OR 'tracheobronchitis'/exp OR bronchitis:ab,ti OR bronchitic:ab,ti
#2	'rhinitis'/exp OR 'allergic rhinitis'/exp OR 'rhinosinusitis'/exp OR 'sinusitis'/exp OR rhinitis:ab,ti OR rhinosinusitis:ab,ti OR sinusitis:ab,ti
#3	'asthma'/exp OR asthma:ab,ti OR asthmatic:ab,ti OR asthmatics:ab,ti
#4	#1 OR #2 OR #3
#5	'antihistaminic agent'/mj OR 'histamine h1 receptor antagonist'/exp OR 'cetirizine'/exp OR 'fexofenadine'/exp OR 'loratadine'/exp OR 'acrivastine'/exp OR 'epinastine'/exp OR 'ebastine'/exp OR 'azelastine'/exp OR 'rupatadine'/exp OR 'desloratadine'/exp OR 'bepotastine'/exp OR 'bilastine'/exp OR 'levocetirizine'/exp OR 'olopatadine'/exp OR 'levocabastine'/exp OR 'mizolastine'/exp OR 'alcaftadine'/exp
#6	"Histamine H1 Antagonist":ab,ti OR "Histamine H1 receptor Antagonist":ab,ti OR "H1RA":ab,ti OR Cetirizine:ab,ti OR Fexofenadine:ab,ti OR Loratadine:ab,ti OR Acrivastine:ab,ti OR Epinastine:ab,ti OR Ebastine:ab,ti OR Azelastine:ab,ti OR Rupatadine:ab,ti OR Desloratadine:ab,ti OR Bilastine:ab,ti OR Levocetirizine:ab,ti OR Olopatadine:ab,ti OR Levocabastine:ab,ti OR Mizolastine:ab,ti OR Alcaftadine:ab,ti
#7	#5 OR #6
#8	#4 AND #7
#9	#9 AND ('crossover procedure'/exp OR 'double blind procedure'/exp OR 'randomized controlled trial'/exp OR 'single blind procedure'/exp OR random* OR factorial* OR crossover* OR 'cross over' OR 'cross-over' OR placebo* OR (doubl* AND blind*) OR (singl* AND blind*) OR assign* OR allocat* OR volunteer*)

Supplementary Table S2. List of non-sedating H1 receptor antihistamines[1] reviewed in this study

Cetirizine, fexofenadine, loratadine, acrivastine, epinastine, ebastine, azelastine, rupatadine, desloratadine, bepotastine, bilastine, levocetirizine, olopatadine Hydrochloride, levocabastine, mizolastine, alcaftadine

Supplementary Table S3. Summary of the participant selection criteria in each clinical trial

Study	Target condition	Inclusion criteria	Exclusion criteria
Allergic asthma			
Gould 1988[2]	Extrinsic allergic asthma	• Reversible airflow limitation (increase of ≥ 15% FEV1 15 minutes after inhalation of 200 µg salbutamol) • Taking oral or inhaled salbutamol or terbutaline • Atopy (positive skin prick test to inhalant allergens including *Dermatophagoides pteronyssinus*, house dust, *Aspergillus fumigatus*, and mixed grass pollens)	• Patients who had taken inhaled or oral corticosteroids, sodium cromoglycate or theophylline derivatives in the three weeks prior to the study
Dirksen 1989[3]	Endogenous or perennial allergic moderate asthma	• More than 20% variation in peak expiratory flow and at least a 15% improvement in FEV1 10 min after inhalation of 0.2 mg salbutamol • Well-controlled asthma with inhaled beclomethasone dipropionate 200 µg b.i.d. with a baseline FEV1 above 65% of the predicted	• Patients who had taken oral steroids in the last 2 months or for longer than 3 months during the last year • Other serious diseases • Pregnant
Allergic rhinitis with asthma			
Grant 1995[4]	Seasonal allergic rhinitis with asthma	• History of seasonal allergic rhinitis that became symptomatic during the same annual pollen season and mild-to-moderate asthma • Documented hypersensitivity to a local seasonal allergen, as determined by skin test or RAST within two years • At least three of the following signs or symptoms of rhinitis: sneezing, nasal congestion, itchy eyes or nasal mucosa, postnasal discharge, watery nasal discharge, or watery eyes • Documented FEV1 between 50% and 80% predicted and at least a 15% improvement from baseline after administration of a bronchodilator	• Symptoms of moderate-to-severe rhinitis or asthma, another concomitant illness that might interfere with the clinical response evaluation • History of intolerance to antihistamines or bronchodilators • History of a neuropsychiatric disorder requiring therapy • History of alcohol or drug abuse • Previous participation in a cetirizine clinical trial, or receipt of an investigational drug within the previous month. • Females who were pregnant, lactating, or not using reliable contraceptive measures • Patients who had used systemic or topical corticosteroids and cromolyn sodium during a week prior to the study, antihistamines for three days prior to

			the study, and astemizole for six weeks prior to the study.
Aaronson 1996[5]	Perennial allergic rhinitis with asthma	• History of perennial asthma • Proven positive skin tests for allergens that correlated clinically with their disease	• Females who were pregnant, lactating, or not using reliable contraceptive measures • Concomitant illness that might interfere with the evaluation of the clinical response, such as acute sinusitis, nasal polyposis, otitis media, upper respiratory infection, or underlying systemic illness; patients with an FEV1 less than 50% predicted at baseline; and any patient with a history of intolerance to antihistamines or bronchodilators. • Patients who had taken inhaled or intranasal steroids or cromolyn sodium during the two weeks prior to the study; those who had taken antihistamines during the 48 hours prior to the study and astemizole for six weeks prior to the study; and those who had taken oral corticosteroids during the week prior to the study
Berger 2002[6]	Seasonal allergic rhinitis with asthma	• A minimum 2-year history of seasonal allergic rhinitis and increased asthma signs or symptoms in conjunction with fall/winter allergy season • Qualified subjects were clinically symptomatic at screening, with moderate nasal rhinorrhea, a total nasal symptom score of at least 6, a total non-nasal score of at least 5, and mild seasonal allergic asthma exacerbations. • FEV1 $\geq$ 70% predicted and a total frequency of asthma score and/or bronchodilator use of a least 2 at the screening visit, and demonstrated reversibility with an inhaled bronchodilator within the past 2 years • Positive IgE-mediated skin test response to relevant allergen within 24 months of screening	• Use of chronic inhaled or systemic corticosteroids • Current or frequent clinically significant upper airway disease, structural nasal abnormalities, and an FEV1 of 70% or less of the predicted normal value. • Use of nasal, oral, or ocular decongestants, nasal topical antihistamines, and corticosteroids
Baena-Cagnani 2003[7]	Seasonal allergic rhinitis with asthma	• At least a 2-year history of SAR and increased asthma symptoms during the autumn allergy season • Clinical symptoms of SAR at the screening and baseline visits	• History of frequent, clinically significant sinusitis; long-term, purulent postnasal drip or rhinitis medicamentosa • History of upper respiratory tract or sinus infection requiring antibiotic therapy within two weeks, or viral

		• FEV1 greater than or equal to 70% of the predicted normal value at the screening visit • Asthma controlled with as-needed bronchodilators only • An increase in FEV1 of at least 12% following bronchodilator inhalation measured within the past 2 years • A total frequency of asthma score and/or bronchodilator use score of at least 2 (indicating at least 'greater than weekly but not daily' symptoms and/or inhaler use) • Positive skin test (prick or intradermal) for seasonal allergen within 24 months of study entry • History of no disease other than SAR and asthma that would affect participation in the study	infection within seven days of screening • Abnormalities of the nasal structure • Ongoing pregnancy or lactation • Requirement for long-term inhaled or systemic corticosteroids; ongoing immunotherapy (unless on a regular maintenance schedule); and dependence on nasal topical antihistamines, nasal corticosteroids, or nasal, oral, or ocular decongestants.
Allergic rhinitis without asthma			
Weiler 1988[8]	Seasonal allergic rhinitis	• History and physical examination findings of seasonal (grass and/or trees) allergic rhinitis for at least two years • Positive skin test or RAST to the appropriate pollen within the past two years	• Women who were of child-bearing potential, pregnant, or lactating • Current symptoms of asthma, receiving immunotherapy, having a clinically significant medical history, or abnormal laboratory values
Ciprandi 1995[9]	Cough-associated allergic rhinoconjunctivitis (*Parietaria Judaica)*	• History of cough associated with allergic rhinoconjunctivitis due to *parietaria judaica* pollen for at least two previous seasons • Positive skin prick tests and RAST for the specific pollen • Normal lung function with FEV1 > 80% predicted	• Patients with prior history of asthma • Patients with previously documented allergy to the studied drug • Women of childbearing potential, and lactating women • Patients who had been treated with any topical or systemic drug in the preceding six weeks • Patients who had received specific immunotherapy before and during the study
NCT00295022	Seasonal allergic rhinitis due to ragweed	• History of seasonal allergic rhinitis due to ragweed for the last two consecutive years • Positive skin test to ragweed pollen • Total Symptom Complex Score ≥18 points	• Any clinically significant condition that might interfere with the treatment evaluation, both for efficacy and safety • Have used forbidden concomitant medications as

		over the combined three half-hourly post-pollen evaluations during priming exposure and over the combined three half-hourly pre-treatment evaluations during the pollen challenge	defined by the protocol
Atopic cough			
Shioya 2004[10]	Atopic cough	• Patients who meet the diagnostic criteria of atopic cough suggested by the Japanese Cough Research Association	• Subjects who had ever taken systemic corticosteroids, cromoglycate, or other anti-allergic agents • Subjects who smoked within the previous eight weeks

FEV1, forced expiratory volume in 1 second; RAST, radioallergosorbent test

Supplementary Table S4. Summary of efficacy endpoints in the 10 included studies

Study	Lung function	Inhalation challenge response	Symptoms	Use of rescue medication
Allergic asthma				
Gould 1988[2]	• FEV1, FVC, RV, TLC (ns) • PEFR (ns) • MMEF (ns) • TGV (ns) • Raw (ns)	• Methacholine PC35 (ns) • Histamine PC35 (↑)	• Cough (ns) • Breathlessness (ns) • Activity limitation (ns) • Sleep disturbance (ns) • Wheeze (↓)	• Salbutamol use (ns)
Dirksen 1989[3]	• FEV1 (ns) • Morning PEF (ns) • Evening PEF (ns)	• Histamine PC20 (↑)	• Cough (ns) • Nocturnal asthma symptoms (ns) • Wheezing (ns) • Dyspnea (ns) • Sputum (ns)	• Salbutamol use (ns)
Allergic rhinitis with asthma				
Grant 1995[4]	• PFT (ns) • Morning PEF (ns) • Evening PEF (ns)		• Cough (↓) • Rhinitis symptoms: sneezing, itching of nose and eyes, postnasal drip, runny nose, and watery eyes (↓), nasal congestion (ns), total rhinitis symptom scores (↓) • Asthma symptoms: chest tightness, wheezing, shortness of breath (↓), sputum production, nocturnal asthma (ns), total asthma symptom scores (↓)	• Albuterol use (ns)
Aaronson 1996[5]	• FEV1, FVC, PEFR, FEF 25-75% (ns) • Mean daily peak expiratory flow rate (ns)	• Methacholine PC20 (ns)	• Cough (ns) • Rhinitis symptoms: itching of nose, nasal congestion, watery eyes (↓), sneezing, itching of eyes, postnasal drip, runny nose (ns), total rhinitis symptom scores (ns) • Asthma symptoms: chest tightness, wheezing, shortness of breath, nocturnal asthma (↓), sputum (ns)	• Albuterol use (ns)

Berger 2002[6]			• Cough (↓) • Rhinitis symptoms: rhinorrhea, nasal congestion, nasal itching, sneezing, itching/burning eyes, tearing/watering eyes, redness of eyes, and itching of ears/palate (↓), total rhinitis symptom scores (↓) • Asthma symptoms: wheezing (↓), difficulty breathing (ns), total asthma symptom scores (ns)	• β2-agonist use (ns)
Baena-Cagnani 2003[7]	• FEV1 (↑), only when baseline FEV1 was less than 80% of predicted normal • Morning PEFR (ns) • Evening PEFR (ns)		• Cough (↓) • Asthma symptoms: wheezing, difficulty breathing (↓), total asthma symptom scores (↓)	• Albuterol use (+)
Allergic rhinitis without asthma				
Weiler 1988[8]			• Cough (ns) • Nose blows, sneeze, nasal itch, sniff (ns) • Postnasal drip, dry nose (ns) • Watery eyes, itchy eyes (ns) • Itchy throat, cough (ns) • Total symptom scores (↓)	
Ciprandi 1995[9]	• PEF, FVC, FEV1 (↑)		• Cough, subjective frequency and intensity (↓) • Conjunctival and nasal symptom score (↓)	
NCT00295022			• Major symptom complex scores (↓) • Runny nose, itchy nose, sniffles, postnasal drip, watery eyes, itchy eyes/ears, itch throat, sneeze, nose blows, nasal congestion (↓) • Cough (ns)	
Atopic cough				
Shioya 2004[10]	• FVC, FEV1 (ns) • Morning PEFR (ns)	• Methacholine (ns) • Cough threshold for	• Cough (↓)	

	• Evening PEFR (ns)	capsaicin (↑)		

FEV1, forced expiratory volume in 1 second; FVC, forced vital capacity; RV, residual volume; TLC, total lung capacity; PEFR, peak expiratory flow rate; MMEF, maximal mid-expiratory flow rate; TGV, total intra-thoracic gas volume; Raw, airways resistance; PEF, peak expiratory flow; PFT, pulmonary function test; FEF, forced mid-expiratory flow rate; Ns, not significant

Arrow indicates a statistical difference in each outcome with active drug (vs. placebo) with the direction of change.

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